

A. Data

- Source and full URL

Philippine Statistics Authority (PSA) - 2022 Annual Poverty Indicators Survey (APIS)
<https://psa.gov.ph/statistics/income-expenditure/apis/node/168406116>

- Description of the data

The dataset comes from Table 1 of the Philippine Statistics Authority's 2022 Annual Poverty Indicators Survey (APIS), which reports the population aged 5 to 24 years by schooling status, for each of the 17 regions of the Philippines (using the region grouping in effect at the time of the survey, including the Bangsamoro Autonomous Region in Muslim Mindanao). For each region, the table gives the total population aged 5-24 and the number of that population currently attending school, both in thousands of persons.

- x and y variables, with units

x (independent variable): Total population aged 5-24 years, in thousands of persons.

y (dependent variable): Population aged 5-24 years currently attending school, in thousands of persons.

- Table of observations

Region	x (thousands)	y (thousands)
NATIONAL CAPITAL REGION	4,900	3,952
CORDILLERA ADMINISTRATIVE REGION	703	593
REGION I - ILOCOS REGION	1,959	1,625
REGION II - CAGAYAN VALLEY	1,364	1,147
REGION III - CENTRAL LUZON	4,605	3,661
REGION IV-A-CALABARZON	6,119	4,943
MIMAROPA REGION	1,317	1,120
REGION V - BICOL REGION	2,563	2,163
REGION VI - WESTERN VISAYAS	2,969	2,376
REGION VII - CENTRAL VISAYAS	3,077	2,490
REGION VIII - EASTERN VISAYAS	1,949	1,652
REGION IX - ZAMBOANGA PENINSULA	1,526	1,170
REGION X - NORTHERN MINDANAO	1,984	1,655

REGION XI - DAVAO REGION	2,093	1,678
REGION XII - SOCCSKSARGEN	2,003	1,653
REGION XIII - CARAGA	1,122	941
BANGSAMORO AUTONOMOUS REGION	1,932	1,508

B. Python Script

```
"""
```

Linear Regression Analysis

Relationship between total population aged 5-24 (x) and population aged 5-24 currently attending school (y), for the 17 Philippine regions.

Model: $y = a_0 + a_1x$

Formulas (least squares, done by hand -- NOT Excel trendline, NOT numpy.polyfit):

$a_1 = [n \cdot \text{Sum}(xy) - \text{Sum}(x) \cdot \text{Sum}(y)] / [n \cdot \text{Sum}(x^2) - (\text{Sum}(x))^2]$

$a_0 = \bar{y} - a_1 \bar{x}$

Requires: openpyxl, matplotlib

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"""
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```
import openpyxl
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```
import matplotlib.pyplot as plt
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```
# CONFIG -- EDIT THESE TO MATCH YOUR EXCEL FILE
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```
FILE_PATH = r"C:\Users\Camille\Downloads\Statistical Tables_0.xlsx"
```

```
# The exact name of the worksheet (tab) that holds Table 1.
```

```
SHEET_NAME = "Table 1"
```

```
# The table is not laid out as one continuous block -- each region's total
```

```
# row is followed by several age-group breakdown rows, and every few
```

```
# regions there's a "Notes / Source" block from the original report.
```

```
# So instead of a start/end range, we list the EXACT row number of each
```

```
# of the 17 regions' TOTAL line (found by inspecting the sheet).
```

```
REGION_ROWS = [
```

```
    23, # NATIONAL CAPITAL REGION
```

```

34, # CORDILLERA ADMINISTRATIVE REGION
45, # REGION I - ILOCOS REGION
56, # REGION II - CAGAYAN VALLEY
86, # REGION III - CENTRAL LUZON
97, # REGION IV-A-CALABARZON
108, # MIMAROPA REGION
119, # REGION V - BICOL REGION
130, # REGION VI - WESTERN VISAYAS
160, # REGION VII - CENTRAL VISAYAS
171, # REGION VIII - EASTERN VISAYAS
182, # REGION IX - ZAMBOANGA PENINSULA
193, # REGION X - NORTHERN MINDANAO
204, # REGION XI - DAVAO REGION
233, # REGION XII - SOCCSKSARGEN
244, # REGION XIII - CARAGA
255, # BANGSAMORO AUTONOMOUS REGION (IN MUSLIM MINDANAO)

```

```
]
```

```
# Column letters (as shown in Excel) for region name, x, and y.
```

```
COL_REGION = "A" # region name
```

```
COL_X = "C"      # Total population aged 5-24 (thousands)
```

```
COL_Y = "D"      # Population aged 5-24 currently attending school (thousands)
```

```
# An x-value NOT among your 17 regions, used for the prediction demo.
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NEW_X_FOR_PREDICTION = 1000 # thousands
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# STEP 1: READ DATA FROM EXCEL
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workbook = openpyxl.load_workbook(FILE_PATH, data_only=True)
```

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sheet = workbook[SHEET_NAME]
```

```
regions = []
```

```
x_values = []
```

```
y_values = []
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```
for row in REGION_ROWS:
```

```
    region_name = sheet[f"{COL_REGION}{row}"].value
```

```
    x_val = sheet[f"{COL_X}{row}"].value
```

```
    y_val = sheet[f"{COL_Y}{row}"].value
```

```

# Skip any row that turned out blank just in case
if x_val is None or y_val is None:
    continue

regions.append(region_name)
x_values.append(float(x_val))
y_values.append(float(y_val))

n = len(x_values)
print(f"Number of regions read from file: {n}")
if n != 17:
    print("WARNING: expected 17 regions -- check REGION_ROWS in CONFIG.")

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# STEP 2: CALCULATE THE SUMS NEEDED FOR THE LEAST-SQUARES FORMULAS
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sum_x = sum(x_values)
sum_y = sum(y_values)
sum_x2 = sum(x ** 2 for x in x_values)
sum_xy = sum(x_values[i] * y_values[i] for i in range(n))

x_bar = sum_x / n
y_bar = sum_y / n

print("\n--- Summary Statistics ---")
print(f"n      = {n}")
print(f"Sum(x) = {sum_x:.4f}")
print(f"Sum(y) = {sum_y:.4f}")
print(f"Sum(x^2)= {sum_x2:.4f}")
print(f"Sum(xy) = {sum_xy:.4f}")
print(f"x_bar  = {x_bar:.4f}")
print(f"y_bar  = {y_bar:.4f}")

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# STEP 3: CALCULATE SLOPE (a1) AND INTERCEPT (a0) USING THE GIVEN FORMULAS

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a1 = (n * sum_xy - sum_x * sum_y) / (n * sum_x2 - sum_x ** 2)
a0 = y_bar - a1 * x_bar

print("\n--- Regression Coefficients ---")
print(f"slope (a1) = {a1:.6f}")
print(f"intercept (a0) = {a0:.6f}")
print(f"\nRegression equation: y = {a0:.4f} + {a1:.4f}x")

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# STEP 4: PREDICTED VALUES AND RESIDUALS
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y_predicted = [a0 + a1 * x for x in x_values]
residuals = [y_values[i] - y_predicted[i] for i in range(n)]

print("\n--- Region-by-Region Results ---")
print(f'{"Region":<20}{x':>10}{y (actual)':>12}{y (pred)':>12}{residual':>12}")
for i in range(n):
    print(f'{str(regions[i]):<20}{x_values[i]:>10.2f}{y_values[i]:>12.2f}"
          f'{y_predicted[i]:>12.2f}{residuals[i]:>12.2f}")

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# STEP 5: ERROR / GOODNESS-OF-FIT MEASURES
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# Sr (also called SSE) = sum of squared residuals = "unexplained" variation
Sr = sum(r ** 2 for r in residuals)

# St = total sum of squares = "total" variation in y around its mean
St = sum((y - y_bar) ** 2 for y in y_values)

# r-squared = fraction of variation in y explained by the regression line
r_squared = (St - Sr) / St

# Standard error of the estimate (syx) -- typical size of a residual

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```
# Divide by (n - 2) because 2 parameters (a0, a1) were estimated
syx = (Sr / (n - 2)) ** 0.5
```

```
print("\n--- Goodness of Fit ---")
print(f"Sr (SSE)          = {Sr:.4f}")
print(f"St (total SS)     = {St:.4f}")
print(f"r^2              = {r_squared:.6f}")
print(f"standard error (Sy/x) = {syx:.4f} (thousand persons)")
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# STEP 6: PREDICTION FOR A NEW x-VALUE (NOT ONE OF THE 17 ORIGINAL POINTS)
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y_new_prediction = a0 + a1 * NEW_X_FOR_PREDICTION
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print("\n--- Prediction for a New x-value ---")
print(f"Using x = {NEW_X_FOR_PREDICTION} thousand (5-24 y/o population)")
print(f"y = {a0:.4f} + {a1:.4f}({NEW_X_FOR_PREDICTION})")
print(f"y = {y_new_prediction:.4f} thousand persons attending school")
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# STEP 7: GRAPH 1 -- SCATTER PLOT WITH REGRESSION LINE
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plt.figure(figsize=(8, 6))
plt.scatter(x_values, y_values, color="blue", label="Actual regional data")
```

```
# Draw the regression line across the range of x
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x_line = [min(x_values), max(x_values)]
```

```
y_line = [a0 + a1 * x for x in x_line]
```

```
plt.plot(x_line, y_line, color="red",
         label=f"Regression line: y = {a0:.2f} + {a1:.4f}x")
```

```
plt.title("Population Aged 5-24 vs. Population Attending School\n(17 Philippine Regions)")
plt.xlabel("Total population aged 5-24 years (thousands)")
plt.ylabel("Population aged 5-24 attending school (thousands)")
plt.legend()
plt.grid(True, alpha=0.3)
```

```
plt.tight_layout()
plt.savefig("scatter_with_regression_line.png", dpi=150)
plt.show()
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# STEP 8: GRAPH 2 -- RESIDUAL PLOT
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```
plt.figure(figsize=(8, 6))
plt.scatter(x_values, residuals, color="green")
plt.axhline(y=0, color="black", linestyle="--") # reference line at 0
```

```
plt.title("Residual Plot")
plt.xlabel("Total population aged 5-24 years (thousands)")
plt.ylabel("Residual = actual y - predicted y (thousands)")
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig("residual_plot.png", dpi=150)
plt.show()
```

C. Regression Results

- Regression equation, slope, and intercept

Regression equation: $y = 40.8742 + 0.7973x$

Slope (a_1) = 0.797254

Intercept (a_0) = 40.874181

- S_r (SSE), r^2 , standard error

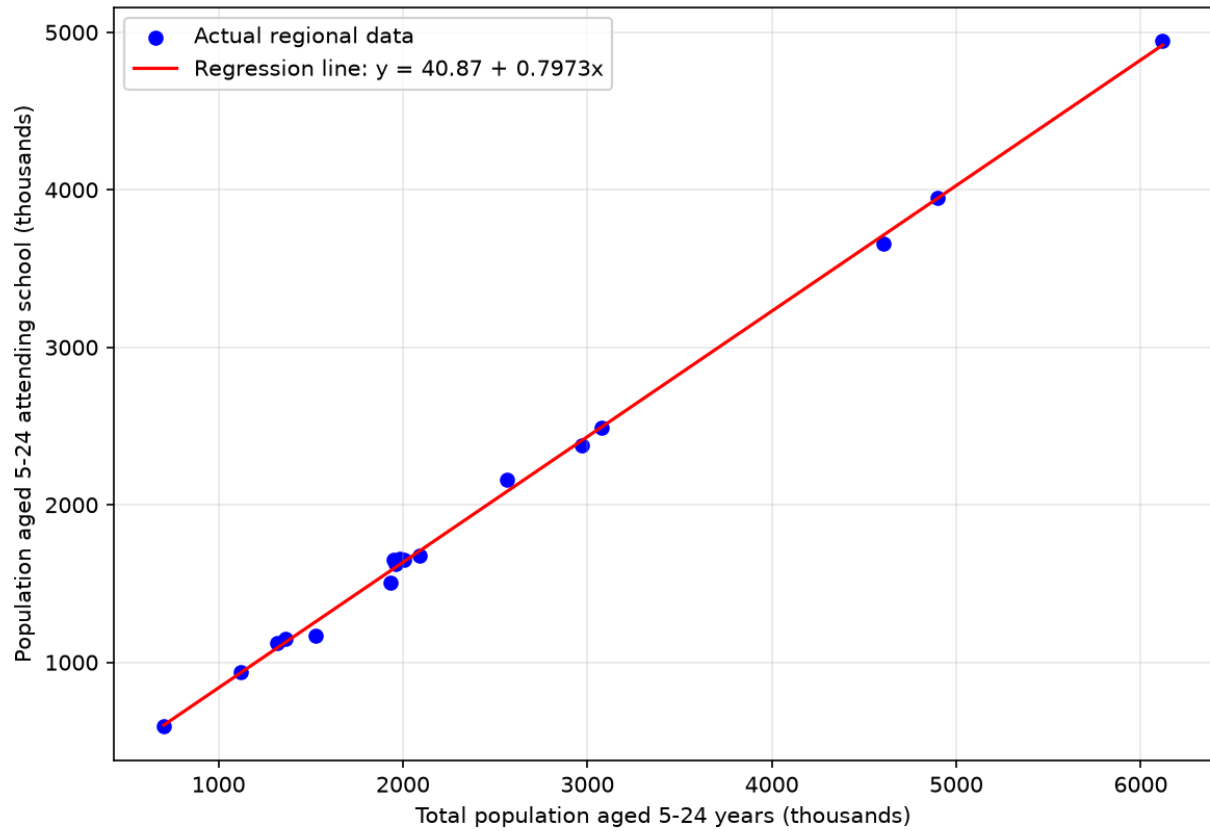
S_r (SSE) = 30805.4932

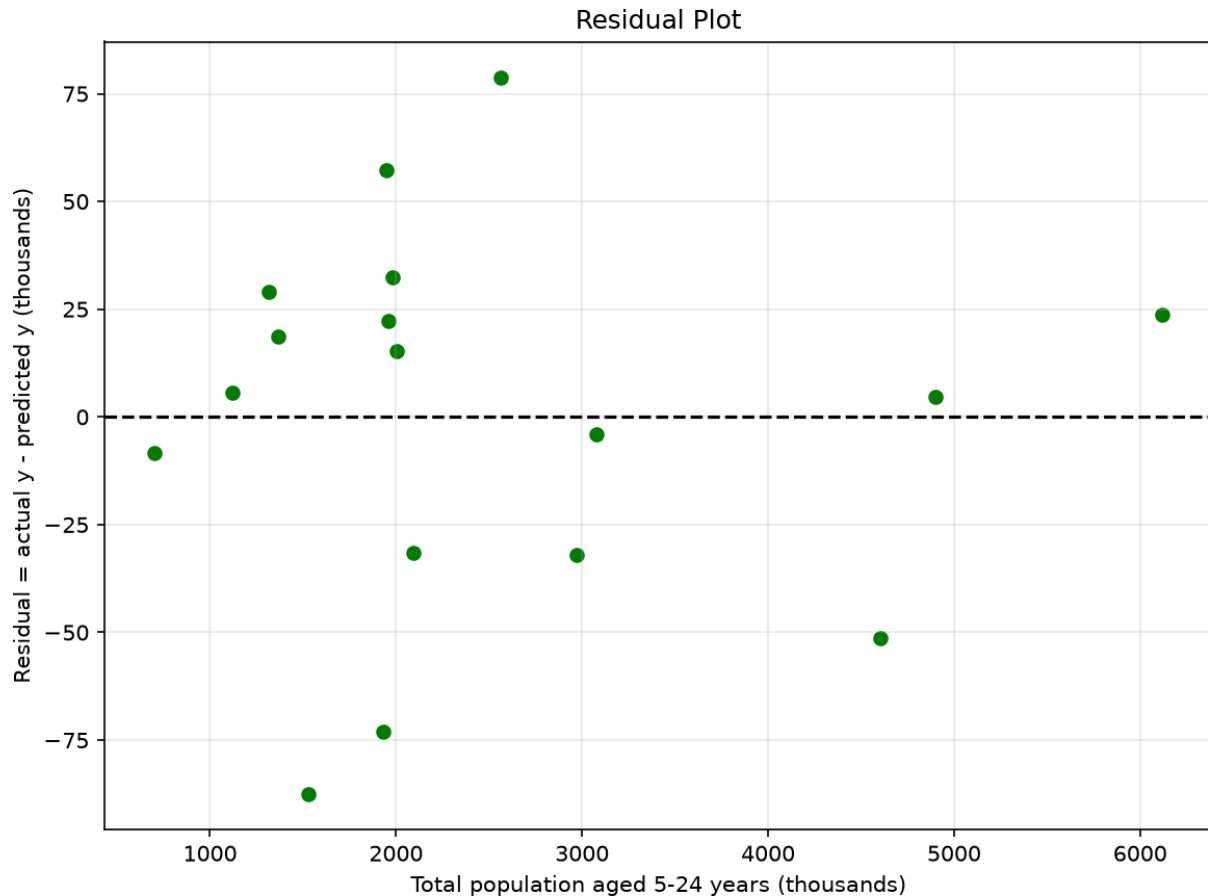
r-squared (r^2) = 0.998585

Standard error of the estimate ($S_{y/x}$) = 45.3178 thousand persons

- Graph with the fitted line, and residual plot

Population Aged 5-24 vs. Population Attending School
(17 Philippine Regions)





- Prediction

For a hypothetical region with $x = 1000$ thousand persons aged 5-24 (a value not among the original 17 regions):

$$y = 40.8742 + 0.7973(1000)$$

$y = 838.1277$ thousand persons predicted to be currently attending school.

- Interpretation

Slope: For every additional 1,000 people aged 5-24 in a region, the model predicts about 797 more of them are currently attending school. The slope represents the rate at which school attendance increases as the youth population increases.

Intercept: The intercept (40.87 thousand) is the model's predicted attendance when $x = 0$. Since no region actually has zero population aged 5-24, this value has little direct real-world meaning; it mainly anchors the line mathematically.

r-squared: An r^2 of 0.9986 means that about 99.86% of the variation in school attendance across regions is explained by variation in the total population aged 5-24. This indicates a very strong linear relationship between the two variables.

Standard error: The standard error of the estimate (45.32 thousand persons) shows the typical size by which the actual attendance figures differ from what the regression line predicts. A smaller standard error means the data points are clustered more tightly around the line.

Residual plot: The residuals are scattered both above and below the zero line without an obvious curve or funnel shape, which suggests that a straight-line model is a reasonable fit for this relationship. Any strong pattern in the residuals would have suggested that a linear model was not appropriate